

V4.1 (9 main actions, 54 sub-actions)

Category	Count	Subcategory Names
1. Grounding	5	Interact with users / physical envs / digital envs / other agents / Augmenting
2. Retrieval	6	events / skills / documents / knowledge / errors / Domain Context
3. Reasoning	12	Generating / Distilling / Inferring / Decomposing / Analysing / Diagnosing / Comparing / Ranking / Contextualizing / Synthesizing / Filtering / Inferencing / Aggregating
4. Learning	8	decision-making / reasoning / grounding / knowledge / LLM parameters / retrieval skills / instructions / Self-improving
5. Planning	3	Decompose / Formulate workflow / Select Directive Strategy
6. Reflection	5	on failures / from in-episode error / with external feedback / through iteration / from memory
7. Evaluate	5	with metrics / with gold / with heuristics / with goals / query
8. Executing	5	strategy / Terminating / debugging / Refusal / anonymity

9. Decision-Making	4	Pick scores / accept or not / multiple candidates / multiple states
Unclassified	12	

● Grounding Actions

Interact with users

Action	Examples & Quotes
Accept instructions from humans	Receive task or command from human user <i>e.g. "book me a flight" or "translate this paragraph"</i> <i>"agent receives task queries and deliverables, completing tasks through multi-turn interactions" (P16)</i>
Ask for clarification from people	Proactively ask human when unclear <i>e.g. "Which file did you mean?"</i>
Inform user of task outcome	Bathing dress recommendation example (P3) <i>Agent informs the user: 'Buy a nice rich navy bathing dress' (P3)</i> WiFi-off notification example (P3) <i>Agent informs the user that wifi is turned off (P3)</i>
Visualize results for human consumption	Generate marker-gene visualization (P1) <i>"CellAgent generated differential expression and marker gene visualizations, enabling intuitive interpretation of cluster identities" (P1)</i>

Format structured output for verification	JSON output format (P2) <i>“generate outputs in a JSON format (or lists of JSON objects)” (P2)</i>
Entertain people	Casual chat / role-play / emotional support <i>Chat casually, tell jokes, do role-play</i>

Interact with physical environments

Action	Examples & Quotes
Affect physical environments via robotic planners	Send language commands to robot arm <i>Move things in the real world by sending language commands to a robot arm</i>
Process perceptual inputs into textual observations	Convert images to text via VLMs <i>VLM image-to-text description</i> Convert sensor / image / audio readings to text <i>Turn images, sounds, or sensor readings into text descriptions for text-based LLMs</i> <i>“Explore skill enhances the LLM-based executor's ability to guide the agent in room exploration by sampling navigation goals from traversable areas ” (P33)</i> <i>“the action Play is utilized to interact with the environment or request re-planning of the current plan S” (P34)</i>

Build semantic map via visual perception models	Use visual perception models, such as depth estimation and instance segmentation, to transform egocentric visual observations into a labeled spatial map with object and layout information. “semantic mapping module receives egocentric visual observations...processed into a depth map and instance segmentation using a UNet and a MaskRCNN” (P34)
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Interact with digital environments

Action	Examples & Quotes
Navigate and interact with web interfaces	<i>“navigate multiple websites, extract information from both structured and unstructured sources” (P2)</i>
Invoke structured API or data endpoint	<i>“Game Control: get_game_state, press_buttons, and navigate_to for direct game control” (P25)</i>
Invoke visual inspection tool	Zoom-in tool for region-of-interest (P4) <i>“proactively invokes the Zoom-in tool for a targeted examination of the specific region of interest” (P4)</i>
Annotate UI Components	Add data-testid attributes (P28) <i>“annotate_interactive_components(file, strategy='add data-testid')” (P28)</i>

Interact with other agents

Action	Examples & Quotes
Conduct multi-agent debate	Multi-agent argumentation toward better answer <i>Have multiple agents argue different sides to reach a better answer</i> <i>"splitting the instruction-following challenge into distinct reasoning and grounding roles handled by a reasoner agent and an actor agent" [P34]</i>
Solve tasks collaboratively across agents	Orchestrator dispatches sub-agents (P25) <i>"A central orchestrator maintains a high-level route plan while dynamically dispatching sub-agents based on game context" (P25)</i> Communicate with peer agents <i>Evaluator provides natural-language feedback (P1)</i> <i>Decompose Development Roles (P28)</i> <i>specifying a scope, or role, for an agent allows it to be treated as a computational object. map each on to an employee in a human organization (P17)</i> <i>"agent engages in multi-turn interactions...to generate deliverables" (P16)</i>
Communicate via Structured Dialogue	Structured chat chain across SDLC phases (P28) <i>"LLM-driven agents communicate via structured chat chains across design, coding, and testing phases" (P28)</i>
Monitor peer agent's decision	Observe another agent's chosen action before making one's own decision — a prerequisite for oversight, override, or trust-based deference in multi-agent systems. (P33) <i>"observes both S_0 and A_0, and decides whether to override the AI's treatment or not"</i>
Override peer agent's recommendation	Replace another agent's selected action with one's own decision — the active side of human-in-the-loop oversight or hierarchical agent control. (P33)

	<i>"If the clinician overrides the AI, the patient outcome Y is determined by S_0 and the alternative treatment H_0 suggested by the clinician"</i>
Recommend action to peer agent	Communicate a proposed action to a peer agent for their consideration, knowing the peer may accept or override it — agent-to-agent advisory communication. (P33) <i>"the AI recommends one of eight possible treatments, which is then reviewed and potentially overridden by the clinician"</i>

Augmenting

Action	Examples & Quotes
Augment external knowledge	Generate Buggy Code Sample via BugRAG (P29) <i>"LLM which generates buggy code by integrating retrieved error slices from BugRAG as context" (P29)</i>
Augment external computation	Execute code <i>Test Runner detects logic inconsistencies (P28)</i> Use calculator / heavy compute tools <i>Call outside tools to handle heavy math</i> <i>run_shell_command tool (425 and 362 invocations, respectively) (P16)</i> <i>"necessitates over 90 precise tool calls, significantly raising the bar" (P16)</i> <i>"deliverables are synced to a Docker-based remote sandbox...emulates human computer operations" (P16)</i>

● Retrieval Actions

Retrieve events

Action	Examples & Quotes
Retrieve events from episodic memory	Importance-weighted retrieval of discoveries (P25) <i>"Persistent memory system storing discoveries (locations, NPCs, items, strategies) with importance-weighted retrieval" (P25)</i> Retrieve Analogical Examples <i>Recall three relevant distinct problems (P28): "Recall three (03) relevant and distinct problems (different from the user task)" (P28)</i> <i>collection and refinement of memories (represented in text) as the system encounters new scenarios (P17)</i>

Retrieve skills

Action	Examples & Quotes
Load skills from the skill library	Grab pre-built skill snippet (Minecraft "chop tree") <i>Find and load ready-made code snippet from pre-built library</i>

Retrieve documents

Action	Examples & Quotes
Extract data from structured and unstructured documents	Read 1–15 documents/tables (P2) <i>"diverse filetype reading; read between 1 to 15 documents and/or tables" (P2)</i>

Leverage documents for code generation	Look up API docs and tutorials <i>Reference docs to help write code</i>
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Retrieve knowledge

Action	Examples & Quotes
Retrieve knowledge from semantic memory	Look up general facts (e.g., HTTP 404) <i>Pull general fact from knowledge base</i>

Retrieve errors

Action	Examples & Quotes
Query Error Repository	Query BugRAG for existing entries (P29) <i>"queries BugRAG to check for existing entries" (P29)</i>

Retrieve Domain Context

Action	Examples & Quotes
Retrieve HLS-related context (P29)	<i>"LLM leverages retrieved HLS-related context to transform input C algorithms or natural language descriptions" (P29)</i>

- Reasoning Actions

Generating

Action	Examples & Quotes
Generate requirements	Auto-generate user-facing requirements (P28) <i>"automatically generates candidate user-facing requirements based on this analysis"</i> (P28)
Propose action candidates	Brainstorm possible next moves <i>Generate one or more possible next moves</i>
Generate initial codes	Cluster semantic codes into themes (P27) <i>"Each agent then emits a set of initial codes; cluster semantically similar codes and generate preliminary themes"</i> (P27)

Distilling

Action	Examples & Quotes
Summarize recent observations and trajectories	Condense flood of info into key takeaways <i>Avoid information overload by summarizing</i>
Distill insights from retrieved information	Binary score with explanation (P3) <i>"For every binary score $z(q)_{i,j}$ from the judge, there is a corresponding explanation $e(q)_{i,j}$"</i> (P3)

Inferring

Action	Examples & Quotes
Infer hidden state from observable evidence	Damage calculation reveals stat distributions (P25) <i>"intelligently infers hidden information through game mechanics: damage calculations reveal stat distributions, move priority ordering constrains speed ranges" (P25)</i>
Infer causal relationship	Determine plausible causal factor (P2) <i>"determine the most plausible event or factor accounting for this variability" (P2)</i>

Decomposing

Action	Examples & Quotes
Decompose into atomic knowledge units	Break character description into atomic knowledge (P22) <i>"break down the given character description into multiple atomic pieces of knowledge" (P22)</i>

Analysing

Action	Examples & Quotes
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Apply Analogical Reasoning	Solutions by analogy to unrelated projects (P28) <i>“generates solutions based on analogies to unrelated projects, which resemble few-shot prompting” (P28)</i>
Analyze codebase structure and behavior	Analyze frontend framework and components (P28) <i>“Analyze frontend framework and interactive components” (P28)</i>
Detect trend in data	Patterns / directions / changes over time (P2) <i>“Identifying patterns, directions, or changes in data over time or across contexts” (P2)</i>
Analyse source code	Read source to understand UI interactions (P28) <i>“analyzes the project’s core functionalities and their interactions with UI elements by reading the source code” (P28)</i>
Understanding intent	<i>“demonstrate clear intent understanding, partial progress” (P5)</i>

Diagnosing

Action	Examples & Quotes
Diagnose Error Cause	LLM analyzes error causes for debug instructions (P29) <i>“The analysis LLM then examines the error causes and provides debugging instructions” (P29)</i>

Comparing

Action	Examples & Quotes
Compare QoR before/after to link directive to hardware (P29)	<i>"Applying UNROLL increases parallelism, which reduces total execution time but requires more logic resources to instantiate parallel hardware units" (P29)</i>
Count and compare values	Quantify and compare across sources (P2) <i>"Quantifying occurrences and comparing values across sources or categories" (P2)</i>

Ranking

Action	Examples & Quotes
Rank items by criteria	Order items by importance (P2) <i>"Ordering items or facts based on specific criteria or importance" (P2)</i>

Contextualizing

Action	Examples & Quotes
Provide context for subsequent LLM calls	Package reasoning as background for next call <i>Feed reasoning so far as context to next LLM call</i>

Synthesizing

Action	Examples & Quotes
Combine info to coherent solution (P2)	<i>"combine information from multiple sources... to produce a coherent solution" (P2)</i>
Correlate variables across sources	Measure variable relationships (P2) <i>"Measuring relationships or associations between two or more variables" (P2)</i>

Filtering

Action	Examples & Quotes
Filter information by threshold	Threshold-based filtering (P2) <i>"Selecting relevant information based on criteria, quality, or thresholds" (P2)</i>

Aggregating

Action	Examples & Quotes
Aggregate multiple candidate outputs	Aggregate tool predictions for cell labels (P1) <i>"aggregates prediction results from multiple tools to generate the final cell type labels" (P1)</i>

Representative value from multi-source data (P2)	<i>"Determining a representative value that summarises numerical data collected from multiple sources" (P2)</i>
Aggregate spatial observations into map (P34)	<i>"supplementary semantic map M^t, which aggregates the information from M_t over successive time steps. The intuition resembles a form of majority voting"</i>

● Learning Actions

Learning decision-making

Action	Examples & Quotes
From random pick to scored selection	<i>Improve the decision process itself</i>

Learning reasoning

Action	Examples & Quotes
Utilize subtask-specific LLMs	One model for summarization, one for SQL <i>Smaller specialized models for specific jobs</i>
Update reasoning via prompt update	Rewrite own prompt template <i>Learn a better way to reason via prompt rewrite</i>

Learning grounding

Action	Examples & Quotes
Update grounding via code-based skills	Improve web-navigation code snippets <i>Write or improve code that interacts with outside world</i>

Learning knowledge

Action	Examples & Quotes
Update the source code as procedural memory	Self-patch agent's own source code <i>Modify the agent's own source code to change behavior</i>
Update semantic memory with knowledge	Expand Error Repository with new mnemonic (P29) <i>"the inspection agent identifies a new error type and integrates the slice into the error repository with a new mnemonic identifier" (P29)</i> <i>"high reliance on web search to offload knowledge retrieval to external sources" (P16)</i>

Learning LLM parameters

Action	Examples & Quotes
Update parametric policy	Train weights for similar tasks <i>Real learning by changing internal weights</i>
Update LLM parameters via SL/RL/RLHF	SL/RL/RLHF weight adjustment <i>Use supervised, reinforcement, or human-feedback learning</i>

Learning retrieval skills

Action	Examples & Quotes
Update retrieval procedures	Better keyword strategies / smarter ranking <i>Improve search and relevance ranking</i>

Learning instructions

Action	Examples & Quotes
Infer instructions from input-output examples	Extract rule from examples for future use <i>Look at input→output examples, figure out underlying rule</i>

Self-improving

Action	Examples & Quotes
Generate-then-train on best outputs	<i>LLM produces multiple outputs, picks best, trains on those — bootstraps improvement</i>
Self-Correct Step Implementation	Modify implementation per error message (P28) <i>"Modify step implementation based on error message" (P28)</i> <i>"leverage feedback to markedly improve performance...feedback-driven self-correction" (P16)</i> <i>"Provide explicit instructions on what must be rectified in the next iteration" (P16)</i>

- Planning Actions

Decompose a task into subtasks

Action	Examples & Quotes
Planner-Executor-Evaluator decomposition (P1)	<i>"Decomposes complex user requests into manageable subtasks, an Executor that carries out the analysis by generating and running code, and an Evaluator that assesses the quality of the results" (P1)</i> <i>"RequireReplan provides the LLM-based executor with the capability to dynamically adjust the plan" (P34)</i>
Plans with sub-steps for navigation (P2)	<i>"formulate plans, decompose problems into sub-steps" (P2)</i>
Subgoal with executable success-condition (P25)	<i>"LLM decomposes the task into a sequence of subgoals, each paired with an executable success-condition function" (P25)</i> <i>delegator can determine which of the k tasks the input string belongs to by observing only a constant number of metadata tokens (P17)</i>
Diverse roles for complex task decomposition (P28)	<i>"assigns diverse roles to multiple agents to efficiently decompose complex tasks" (P28)</i>
Decompose Task into Subproblems (P28)	<i>"Create JavaScript functionality handling empty display and consecutive operator clicks" (P28)</i> <i>"LLM-based planner to decompose the specified language instruction L into a sequence of subtasks $S = [S_0, S_1, \dots, S_n]$" (P34)</i> <i>"Decomposing hard problems into subproblems often makes them easier and more efficient to solve" (P17)</i> <i>Specialists each solve a distinct task type, receiving only the relevant portion of the input (P17)</i>

Formulate a workflow from task structure

Action	Examples & Quotes
Comprehensive analysis workflow (P1)	<i>"the Planner accurately interprets user intent and formulates a comprehensive analysis workflow" (P1)</i>
Multi-step web navigation plan (P2)	<i>"planning to navigate the web; tasks require navigation through an average of 4.2 web pages" (P2)</i>
HLSTuner directive insertion plan (P29)	<i>"HLSTuner formulates a detailed plan that specifies: (1) the combination of HLS directives, (2) target code segments, and (3) the insertion actions" (P29)</i>
Plan Code Structure: HTML/CSS/JS (P28)	<i>"Planning: HTML Structure, CSS Styling, JavaScript Functionality" (P28)</i>

Select Directive Strategy

Action	Examples & Quotes
HLS directive combinations on loops/arrays (P29)	<i>"selects effective HLS directive combination strategies and inserts directives within the specific structure (e.g., loops and arrays)" (P29)</i>

● Reflection Actions

Reflecting on failures (overall goal)

Action	Examples & Quotes
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Analyze stuck/failed state vs ground truth (P25)	<i>"Analyzes stuck states by comparing current situation against ground truth sources (porymap data, knowledge base) to diagnose navigation failures" (P25)</i>
Debug Test Logic from failure log (P28)	<i>"Modify test script based on failure log" (P28)</i>
Align Requirements to validated tests (P28)	<i>"Refine requirement based on validated test cases to ensure alignment" (P28)</i>
Wrong conclusion despite correct info (P2)	<i>"When the agent reaches an incorrect conclusion despite accessing the correct information, due to flaws in logical reasoning, interpretation, or multi-step analytical processes" (P2)</i>
Self-Correct Step Implementation (P28)	<i>"Modify step implementation based on error message" (P28)</i> <i>debugging specialist rewrites the entire code every time it fixes a bug (P17)</i>
Explicit modification instructions (P29)	<i>"This model formulates explicit modification instructions with detailed analysis" (P29)</i>
Reflect on failed episodes (P34)	<i>"RequireReplan...dynamically adjust the plan, improving the robustness to exceptions"</i> <i>"the grounded prior knowledge prevents the agents from repetitive errors and facilitates grounded exception handling" (P34)</i> <i>"Explain the failure from a user requirement perspective" (P16)</i>

Reflecting from in-episode error (in execution)

Action	Examples & Quotes
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Navigation error recovery (P2)	<i>“Navigation errors — when the agent fails to locate or access the correct source of information” (P2)</i>
Synthesis error recovery (P2)	<i>“Synthesis Error — when the agent reaches an incorrect conclusion despite accessing the correct information” (P2)</i>
Executor self-corrects on execution error (P1)	<i>“In case of an execution error $E(ci)$, the Executor autonomously performs self-correction to produce a valid code version” (P1)</i>
Analyse log to formulate error modification actions	Reasoning-to-instruction on HLS log (P29) <i>“an analysis agent adopts a reasoning-to-instruction method, analyzing the HLS log to formulate error modification actions” (P29)</i>
Inspect Error Pattern in HLS log (P29)	<i>“an inspection agent examines the erroneous code and error messages parsed from the HLS tool test results” (P29)</i>
Reflect on Error Fix for reasonableness (P29)	<i>“reflects on the code after assuming the fix to ensure the modification is reasonable” (P29)</i>

Reflecting with external feedback

Action	Examples & Quotes
Adjust action parameters based on feedback	Auto-adjust params from exception info (P1) <i>“automatically adjusts parameters using exception information to generate executable code” (P1)</i>

Revise output based on evaluator feedback	Iterative theme refinement (P27) <i>“feedback from the feedback agent is used to iteratively refine and improve the generated themes” (P27)</i>
Simulate grounding feedback internally	Mental rehearsal before execution <i>Imagine “if I do X, what will the environment look like?” without executing</i>
Incorporate feedback from another agent	Evaluator natural-language feedback (P1) <i>“the Evaluator agent ALLMe assesses the outcome and provides natural language feedback if it deems revisions are necessary” (P1)</i>
Correct reasoning direction mid-trajectory	Update hypothesis with new visual evidence (P4) <i>“allowing the model to update its textual understanding, refine its hypothesis, or even trigger further visual exploration” (P4)</i>

Reflecting through iteration

Action	Examples & Quotes
Refine outcome over multiple rounds (self-reflective)	Self-reflective optimization mechanism (P1) <i>“The Evaluator drives a self-reflective optimization mechanism, which leverages automated evaluation methods... to iteratively refine outcomes” (P1)</i>

Refine Directive Strategy Iteratively	Iterative refinement using current QoR (P29) <i>“When the initial attempt fails... HLSTuner activates an iterative refinement incorporating current directives and the resulting QoR” (P29)</i>
Iterative precision enhancement loop (P1)	<i>“This self-reflective optimization loop iterates to enhance precision, and the final results from all subtasks are synthesized” (P1)</i>
Iterate toward a quality threshold	Three iterations with different algorithms (P1) <i>“the optimization process runs three iterations, each invoking a different algorithm” (P1)</i>
Simulate grounding feedback internally	Mentally simulate an alternative scenario — either before acting (imagine 'if I do X, what happens?') or post-hoc (imagine 'what would have happened if action X had been different?'). Includes counterfactual trajectory generation for causal analysis. (P33) <i>"analyzing this effect necessitates counterfactual reasoning across three distinct scenarios"</i>

Reflecting from memory

Action	Examples & Quotes
Learn from mistakes in working memory	Real-time learning before memory discard (P1) <i>“This allows the agent to learn from mistakes in realtime and avoid repeating errors before the local memory is discarded” (P1)</i>

Reflect through self-monitoring

Action	Examples & Quotes
Self-monitoring	Real-time learning before memory discard (P1) <i>“These steps are labeled as reflective, indicating the model is engaging in self-monitoring or explicit reasoning before issuing commands” (P1)</i>

- Evaluate Actions

Evaluating with metrics

Action	Examples & Quotes
Halve parallelism if utilization exceeds budget (P29)	<i>“HLSTuner analyzes QoR to scale loop parallelism up or down. For example, if hardware utilization exceeds the budget, HLSTuner halves parallelism” (P29)</i>
Reproducibility score 1–4 (P11)	<i>“In Phase 2, they inspect the provided code for potential inconsistencies; generating a reproducibility score on a scale from 1 to 4” (P11)</i>
Trustworthiness scores $s^{\wedge}(t) = (C, D, T)$ (P27)	<i>“Evaluation Scores: Obtain $s^{\wedge}(t) = (C, D, T)$ on the trustworthiness dimensions” (P27)</i>

Score Repair Candidates	Judge selects optimal repair suggestion (P29) <i>“the scoring agent (functioning as the judge) based on clarity, logical soundness, alignment with error messages, this agent selects the optimal suggestion” (P29)</i>
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Evaluating with gold

Action	Examples & Quotes
Review Code Against Golden Standard	Pair buggy segments with errors for CoT (P29) <i>“we prompt LLM to review the correct HLS-C code, pairing buggy code segments with corresponding error messages to construct debugging CoT” (P29)</i> <i>"Evaluates code quality and logic based on the text deliverables and rubrics" (P16)</i>
HLSFixer retest vs golden results (P29)	<i>“HLSFixer retests the corrected HLS design against the golden results to ensure semantic equivalence” (P29)</i>
Gold Checker: equivalence/completeness/correctness (P28)	<i>“The Gold Checker evaluates each annotation on three binary criteria: equivalence, completeness, and correctness, where a score of 1 indicates the criterion is met” (P28)</i>

Evaluating with heuristics

Action	Examples & Quotes
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Reduce trivial-bug generation (P29)	<i>"The agent assesses the contextual applicability of potential bugs, reducing the probability that the LLM forcibly generates trivial results" (P29)</i> <i>"Detect logical errors, missing functionalities, or violations of best practices" (P16)</i>
Verify intermediate results	Domain-expert agent verification (P28) <i>"agents with domain expertise verify intermediate results and reduce errors" (P28)</i>
Quality control before normalization (P1)	<i>"codified best-practices, such as the standard order of operations (e.g., quality control must precede normalization)" (P1)</i>
Recognize Knowledge Boundary	Identify queries outside role scope (P22) <i>"recognize and refuse queries that conflict with their role knowledge" (P22)</i>
Verify output accuracy via consistency check	Theme-quote consistency check (P27) <i>"task an evaluator agent with determining whether each theme is consistent with its supporting quotes" (P27)</i> <i>"list exactly which rubrics were not met" (P16)</i>
LLM-as-judge / heuristics (P22)	<i>"we use GPT-4o as the default evaluator, each dimension scored on a scale of 0 to 2" (P22)</i> <i>"map the agent's deliverables to a score ranging from 0 to 10" (P16)</i> <i>"Evaluates dynamic behavior and visual correctness based on screenshots" (P16)</i> <i>"Analyze {Visual Assets} for UI elements, layout consistency, and text rendering" (P16)</i> <i>"Verify the implementation of every item listed in {Rubrics}" (P16)</i>

Evaluating with goals

Action	Examples & Quotes
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Combine subtasks to meet user requirements (P1)	<i>“the final results from all subtasks are synthesized to meet the user’s requirements” (P1)</i>
Independent goal-completion check (P25)	<i>“Independently checks whether objectives are truly complete, preventing the orchestrator from advancing when the main agent incorrectly believes a task is finished” (P25)</i>
Detect query-role conflict (P22)	<i>“violates specific content within the role profile” (P22)</i>
QoR-aware reasoning aligns with hardware (P29)	<i>“HLSTuner enables QoR-aware reasoning to align optimization goals with hardware constraints” (P29)</i>

Evaluating query

Action	Examples & Quotes
Acknowledge False Information	Surface false premises in query (P22) <i>“Evaluating whether the model recognizes potential errors in the query” (P22)</i>

● Executing

Executing strategy

Action	Examples & Quotes
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Insertion agent executes HLS-C optimization (P29)	<i>"an insertion agent executes this plan for HLS-C optimization" (P29)</i>
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Terminating

Action	Examples & Quotes
Terminate rollout with answer tags (P4)	<i>"produces a final answer and terminates the rollout; enclose it within <answer></answer> tags" (P4)</i>

Execute debugging

Action	Examples & Quotes
Strict-instruction debugging implementation (P29)	<i>"Operating under strict instruction adherence, this agent adopts the instructions to implement debugging" (P29)</i> <i>M_q has the capacity to identify exactly one bug at a time (P17)</i>

Generate Refusal Response

Action	Examples & Quotes
Character-consistent refusal with explanation (P22)	<i>"providing clear refusal responses with appropriate explanations" (P22)</i>

Executing with anonymity

Action	Examples & Quotes
Mask algorithm IDs with generic labels (P1)	<i>"all algorithm identifiers are masked before evaluation ... replaced by generic labels" (P1)</i>

● Decision-Making Actions

Make a decision

Action	Examples & Quotes
Make a decision according to memory	<i>The agent conditions its decisions on I, the current observation st, and a history buffer of n past state-action pairs $P5$</i>

Pick scores

Action	Examples & Quotes
argmax / softmax / majority vote	<i>argmax = highest score; softmax = probabilistic sample; majority vote across evaluators</i>

Decide accept or not

Action	Examples & Quotes
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Decline out-of-scope queries (P22)	<i>“appropriately reject queries that exceed their knowledge boundaries or conflict with their role settings” (P22)</i>
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Generate multiple candidates

Action	Examples & Quotes
Multifaceted debugging instruction generation (P29)	<i>“LLM Group for multifaceted evaluation generates diverse debugging instructions” (P29)</i>
Running multiple candidate pipelines (P1)	<i>“Running multiple candidate pipelines for each analysis step” (P1)</i>

Decide multiple states

Action	Examples & Quotes
Fork generative state at uncertainty point (P4)	<i>“Fork the current generative state and generate a new parallel trajectory from this point of high uncertainty” (P4)</i>

● Memory Actions

Store Information

Action	Examples & Quotes
Store information in working memory	Scratchpad for current inputs and intermediate results <i>Quick-access buffer that holds recent inputs and intermediate results</i> <i>todo_write tool (15 invocations by GPT-5.2) (P16)</i> <i>"Gemini-3-Pro...initialize_memory_bank (7 times) "Gemini-3-Pro...update_memory_bank (22 times)...attempts to persist state and key information externally"" (P16)</i>
Store episodic trajectories	Record full action sequences for later training <i>Save complete sequences from start to finish for training/review</i>
Store knowledge in semantic memory	Save general world facts <i>Save general facts about the world, not tied to any specific event</i>
Store experiences in episodic memory	Save events as episodes <i>Saves past events as personal-diary-like episodes (which game won, which plan failed)</i>

Discard information

Action	Examples & Quotes
Discard information from working memory	Discard local memory after subtask completion (P1) <i>"the local memory is discarded upon successful completion of the subtask" (P1)</i>

Convert memory

Action	Examples & Quotes
Consolidate working memory into long-term memory	Retain only successful analysis path to global memory (P1) <i>Discard-after-use strategy retains only the clear and successful analysis path, preventing trial-and-error interference (P1)</i>

Read memory

Action	Examples & Quotes
Read from working memory	Grab intermediate results from scratchpad <i>Check the scratchpad — grab stored intermediate results and state</i>

● Unclassified

Items pending categorization. Add entries below as they come up in discussion.

Maintain Cross-File Consistency

Action	Examples & Quotes
Share project context for non-conflicting test-ids (P28)	<i>"Agents share project context to ensure: Cross-file consistency; Non-conflicting test-ids" (P28)</i>

Compact context window

Action	Examples & Quotes
Discard redundant game state, keep summaries (P25)	<i>“automatic context compaction to manage thousands of reasoning steps, preserving only LLM responses and action summaries while discarding redundant game state” (P25)</i>

Maintain curriculum library

Action	Examples & Quotes
Skill syllabus arranged by difficulty	<i>Keep mastered + upcoming skills in difficulty order</i>

Establish a non-parametric policy

Action	Examples & Quotes
Build case library, look up nearest match	<i>Instead of weight changes, build case library of past experiences</i>

Build semantic map via VLM

Action	Examples & Quotes
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Annotated environment map ("door here", "enemy there")	<i>VLM scans environment, produces labeled map with annotations</i>
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Role Conditioning Actions

Adopt Expert Persona

Action	Examples & Quotes
Role-condition GPT-4o pool for interview analysis (P27)	<i>"A pool of $k=4$ role-conditioned GPT-4o agents are each given the full interview transcript as input" (P27)</i>

Simulate human persona

Action	Examples & Quotes
Multi-behavior network from simulated reposts (P21)	<i>"global interactive multi-behavior overlapping network is constructed based on the simulated behaviors of reposting and reposting with a comment, as well as following" (P21)</i>
Reader-agent prompt template (P21)	<i>"construct prompt templates for creating reader agents based on user attributes ua and user historical posts" (P21)</i>
Role-play personality imitation (P22)	<i>"I want you to play as {role}, imitating {role}'s personality and values" (P22)</i>

Maintain role even on refusal (P22)	<i>"maintain its role-playing ability even when refusing to answer" (P22)</i>
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Retain Role-Specific Perspective

Action	Examples & Quotes
Keep domain lens for diverse theme formulation (P27)	<i>"others may retain role-specific perspectives to support diverse theme formulation" (P27)</i>

V3 (validated):

● Memory Actions

Action	Plain Explanation	Evidence
Store information in working memory	Put whatever the agent is currently using — recent inputs, intermediate results — onto a "scratch pad" for quick access.	
Store/update experiences in episodic memory	Save past events as "episodes" — which game was won, which plan failed, what happened step by step — like a personal diary.	